This can be written in matrix form as

$$\begin{bmatrix} a_{11}\, a_{12} \ldots . a_{1n} \\ a_{21}\, a_{22} \ldots a_{2n} \\ \ldots\ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots\ldots \\ a_{m1}\, a_{m2} \ldots a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ . \\ . \\ b_m \end{bmatrix}$$

Thus we get the matrix equation

$AX = B$ … (1)

where $A = \begin{bmatrix} a_{11}\, a_{12} \ldots . a_{1n} \\ a_{21}\, a_{22} \ldots . a_{2n} \\ .. \ldots .... \\ .. \ldots .... \\ a_{m1}\, a_{m2} \ldots . a_{mn} \end{bmatrix}$;

$X = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ . \\ . \\ b_m \end{bmatrix}$

From (1) solution is

$$\boxed{X = A^{-1}B}$$

Example 1.20

Solve by using matrix inversion method:

$2x + 5y = 1$

$3x + 2y = 7$

Solution

The given system can be written as

$$\begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

i.e., $AX = B$

$X = A^{-1}B$

where $A = \begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$

$$|A| = \begin{vmatrix} 2 & 5 \\ 3 & 2 \end{vmatrix}$$

$= -11 \neq 0$

A^{-1} exists.

$$adj\ A = \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} adjA$$

$$= \frac{1}{-11} \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix}$$

$X = A^{-1}B$

$$= \frac{1}{-11} \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

$$= -\frac{1}{11} \begin{bmatrix} -33 \\ 11 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$x = 3$ and $y = -1$.

Example 1.21

Solve by using matrix inversion method:

$3x - 2y + 3z = 8;\ 2x + y - z = 1;$

$4x - 3y + 2z = 4$

Solution

The given system can be written as

$$\begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$$

i.e., $AX = B$

$X = A^{-1}B$

Here $A = \begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and

$B = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$

$$|A| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{vmatrix}$$

$= -17 \neq 0$

01_11th_BM-STAT_Ch-1-EM.indd 13 21-04-2020 12:20:04 PM